

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO4:

Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards

(BL2,BL6)

Assessment Tool Weightage: 30%

Annexure A

Industry Problem-Solving Task

1. A star topology is ideal for a 20-person startup because all devices connect to a central switch, making the network easy to manage and troubleshoot. At the core sits a Layer 2/Layer 3 managed switch (such as a Cisco Catalyst or Netgear ProSafe) with at least 24 ports to accommodate all workstations, printers, and VoIP phones with room to grow. Each device connects to the switch using Cat6 UTP cables, which support Gigabit speeds up to 100 meters — more than sufficient for any typical office floor plan. The switch uplinks to a router/firewall (such as a Cisco RV series or pfSense appliance) that handles internet access, DHCP, and network address translation. A Wi-Fi access point connected to the switch extends wireless coverage for laptops and mobile devices. The key advantage of this design is fault isolation: if one workstation's cable fails, only that single node goes offline, leaving all other 19 employees unaffected. Centralised cable management also simplifies future moves, adds, and changes.
2. A bank demands near-zero downtime, making a full or partial mesh topology the right choice. In a full mesh, every core network device — routers, core switches, and servers — connects directly to every other device, so no single link or node failure can bring

down the network. For a bank, a partial mesh is typically deployed at the core layer (connecting all core switches and firewalls to each other) while the distribution and access layers use redundant uplinks to at least two core switches. Dual enterprise-grade routers (such as Cisco ASR or Juniper MX series) provide BGP-based redundant internet connectivity. All links between core devices use fiber optic cables — either single-mode for long distances between buildings or multi-mode OM4 within the data center — because fiber is immune to electromagnetic interference, supports very high bandwidth, and is essential for maintaining low-latency financial transactions. Redundant power supplies, UPS systems, and RAID storage arrays further complement the mesh design. The justification is straightforward: when a link fails in a mesh, traffic automatically reroutes through alternative paths in milliseconds using protocols like OSPF or BGP, satisfying regulatory requirements for uptime and data integrity.

3. For a small classroom lab with a tight budget, bus topology offers the lowest hardware cost because all computers share a single common communication line — the bus — eliminating the need for a central switch or hub. A single coaxial cable (RG-58, 50-ohm) or, in a modern variation, a single Cat5e/Cat6 cable daisy-chained through an inexpensive unmanaged hub runs along the length of the room. Each computer connects to the backbone using a T-connector and a BNC connector in a classic coaxial setup, with terminators (50-ohm) placed at both ends of the cable to absorb signals and prevent reflection. For a more contemporary low-cost implementation, a single inexpensive 8- or 16-port unmanaged switch replaces the terminators and uses Cat5e UTP patch cables to each workstation — achieving bus-like simplicity at minimal cost. The primary limitation is that the network is susceptible to collisions and a single cable break disrupts all connected machines, but for a supervised classroom environment with light usage (internet browsing, file sharing), this trade-off is entirely acceptable. No dedicated server is required; a teacher's PC can share resources using built-in Windows/Linux file sharing.
4. A large university campus serves thousands of users across multiple buildings — lecture halls, labs, libraries, dormitories, and administration — making a single topology inadequate. A hybrid topology combines the best elements of multiple designs: a fiber-based ring or mesh at the campus backbone level for redundancy, a star topology within each building for manageability, and bus or tree structures within individual labs for cost-efficiency. At the campus core, high-speed fiber optic cables (single-mode, 10 Gbps or

40 Gbps) interconnect core routers and multilayer switches using a ring or partial mesh arrangement, ensuring that no single link failure isolates an entire building. Each building has its own distribution switch connected to the campus core via dual fiber uplinks. Within each building, a traditional star topology branches from the distribution switch to floor-level access switches, which then connect individual devices via Cat6 cables. This hierarchical three-layer model — core, distribution, and access — is the industry standard for enterprise and campus networks because adding a new building simply means installing a new distribution switch and running fiber to the core, with no disruption to existing infrastructure. Wireless access points throughout the campus connect via PoE (Power over Ethernet) to access switches, eliminating the need for separate power cabling. The hybrid design thus delivers redundancy at the backbone, simplicity at the building level, and scalability campus-wide.

5. Designing a network for a company with multiple departments involves creating a structured and scalable architecture using switches and routers to ensure efficient communication and security. Each department (such as HR, Finance, Sales, and IT) is connected through dedicated Layer 2 switches that form the access layer, allowing devices like computers and printers to communicate within the same department. These switches are then connected to a central Layer 3 switch or router, which acts as the core layer and enables communication between different departments through inter-VLAN routing. Virtual LANs (VLANs) are configured to logically separate each department's network, improving security and reducing broadcast traffic. A router is also used to connect the internal network to external networks such as the internet, often with firewall functionality for security. Redundant links and backup paths can be included to ensure reliability, while proper IP addressing and subnetting are applied for efficient network management. This hierarchical design improves performance, enhances security, and allows easy expansion as the company grows.

2.A bank demands near-zero downtime, making a full or partial mesh topology the right choice. In a full mesh, every core network device — routers, core switches, and servers— connects directly to every other device, so no single link or node failure can bringdown the network. For a bank, a partial mesh is typically deployed at the core layer (connecting all core switches and firewalls to each other) while the distribution and access layers use redundant uplinks to at least two core switches. Dual enterprise-grade routers (such as Cisco ASR or Juniper MX series) provide BGP-based redundant internet connectivity. All links between core devices use fiber optic cables — either single-mode for long distances between buildings or multi-mode OM4 within the data center — because fiber is immune to electromagnetic interference, supports very high bandwidth, and is essential for maintaining low-latency financial transactions. Redundant power supplies, UPS systems, and RAID storage arrays further complement the mesh design. The justification is straightforward: when a link fails in a mesh, traffic automatically reroutes through alternative paths in milliseconds using protocols like OSPF or BGP, satisfying regulatory requirements for uptime and data integrity.

Solution:

1. Introduction

This report describes the implementation of the network topology in Cisco Packet Tracer. The network consists of two routers (R1 and R2), four switches (SW1, SW2, SW3 and SW4), and four PCs (PC1, PC2, PC3 and PC4).

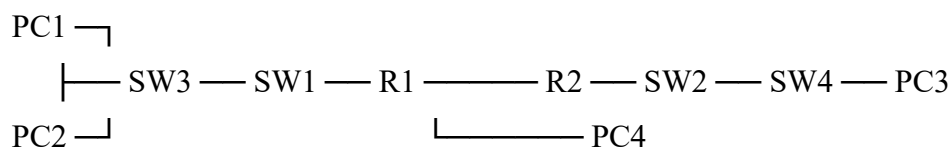
IP addressing was configured on the PCs and router interfaces. The serial connection between R1 and R2 was configured and activated. OSPF was then configured to provide dynamic routing between the two LAN networks.

2. Network Topology

The topology contains two LAN networks connected through two routers.

- **Left LAN:** 192.168.10.0/24
- **Right LAN:** 192.168.20.0/24
- **R1–R2 serial network:** 10.0.0.0/30

The devices are connected as follows:



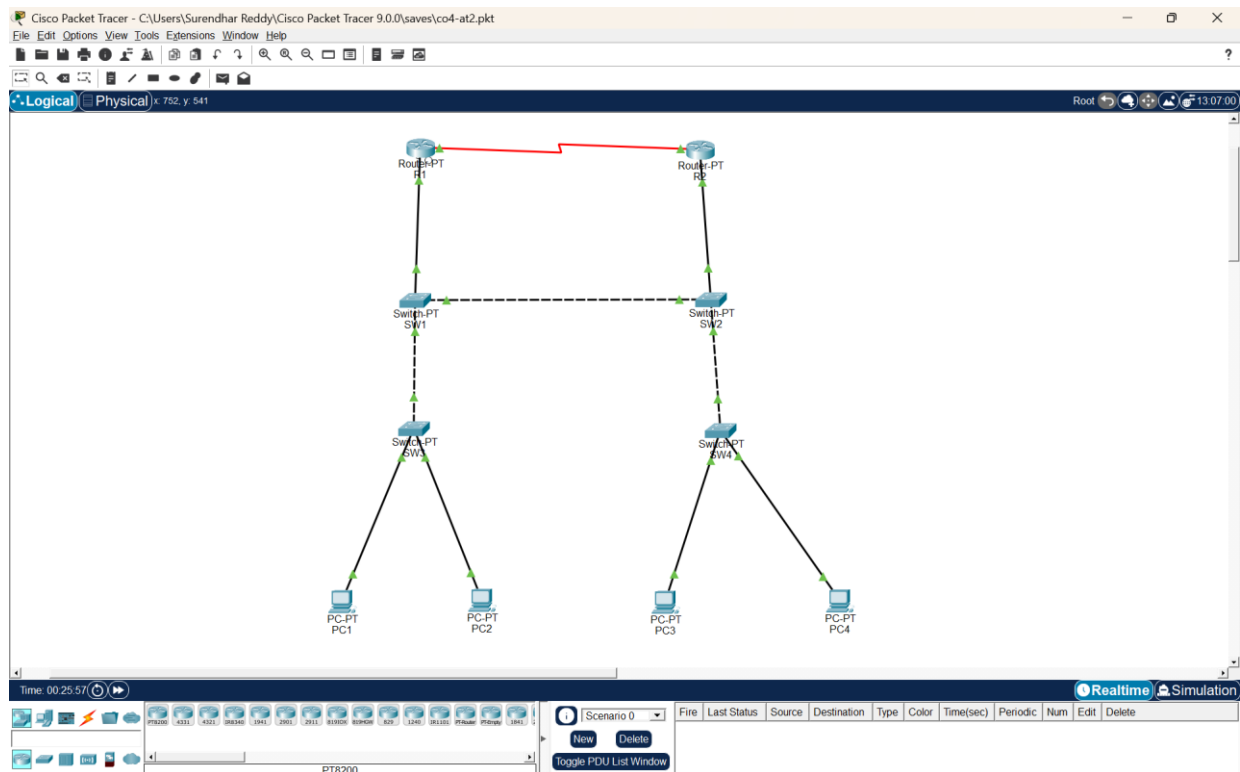


Figure 1: Complete network topology in Cisco Packet Tracer

3. IP Addressing

Device	Interface	IP Address	Subnet Mask
R1	FastEthernet0/0	192.168.10.1	255.255.255.0
R1	Serial2/0	10.0.0.1	255.255.255.252
R2	FastEthernet0/0	192.168.20.1	255.255.255.0
R2	Serial2/0	10.0.0.2	255.255.255.252
PC1	NIC	192.168.10.10	255.255.255.0
PC2	NIC	192.168.10.11	255.255.255.0
PC3	NIC	192.168.20.10	255.255.255.0
PC4	NIC	192.168.20.11	255.255.255.0

Default Gateways

- PC1 → 192.168.10.1
- PC2 → 192.168.10.1
- PC3 → 192.168.20.1
- PC4 → 192.168.20.1

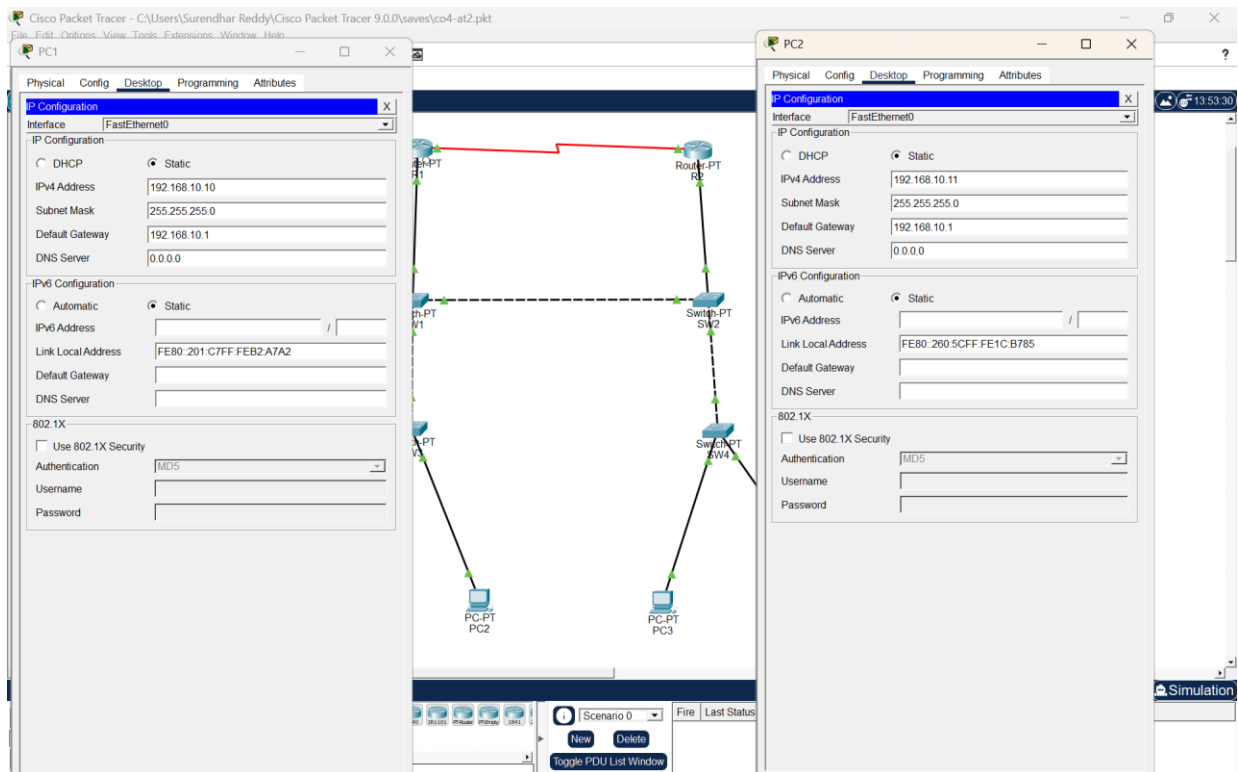


Figure 2: IP configuration of PC1 and PC2

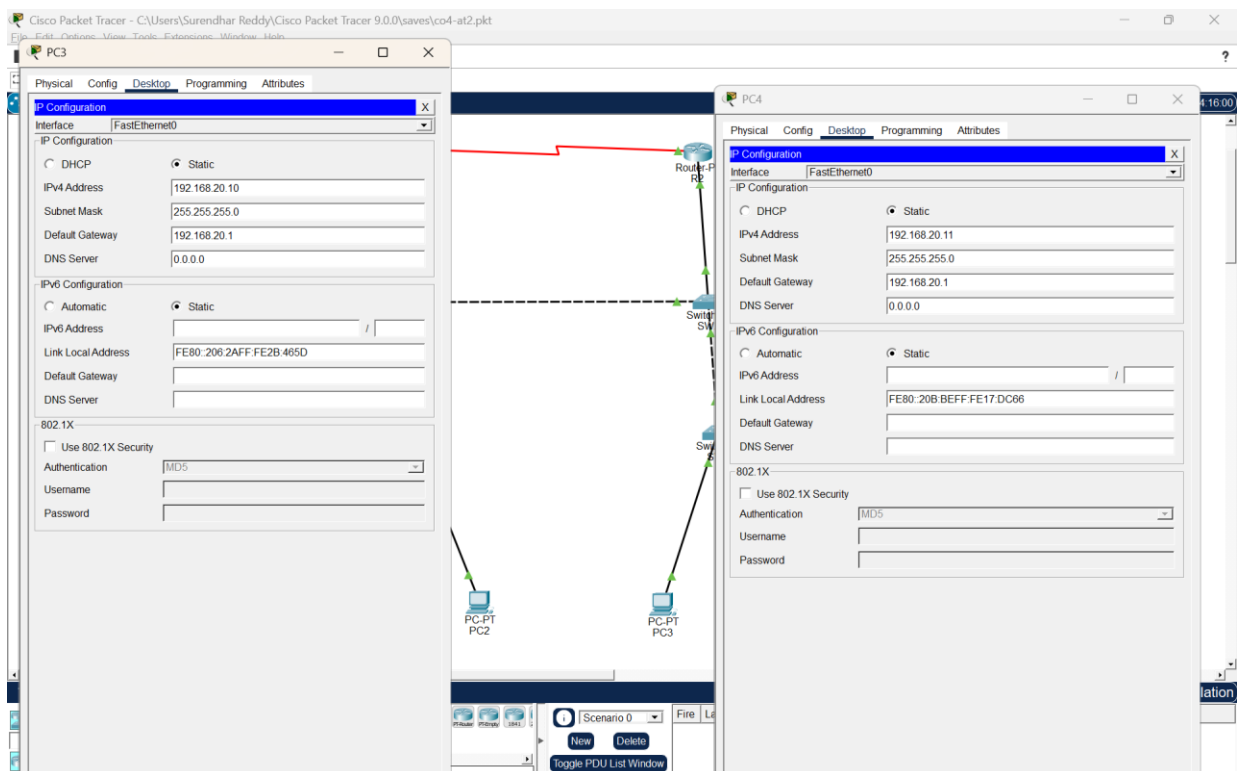


Figure 3: IP configuration of PC3 and PC4

no shutdown

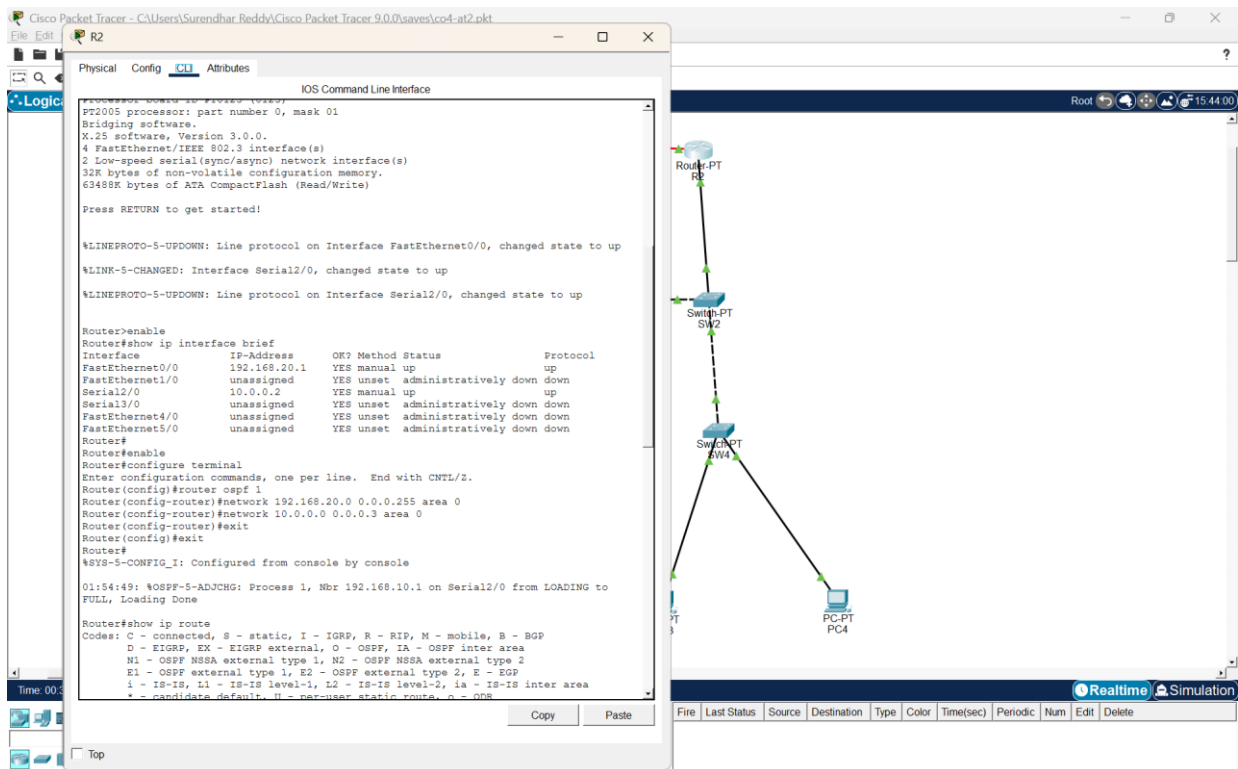


Figure 5: R2 interface IP configuration

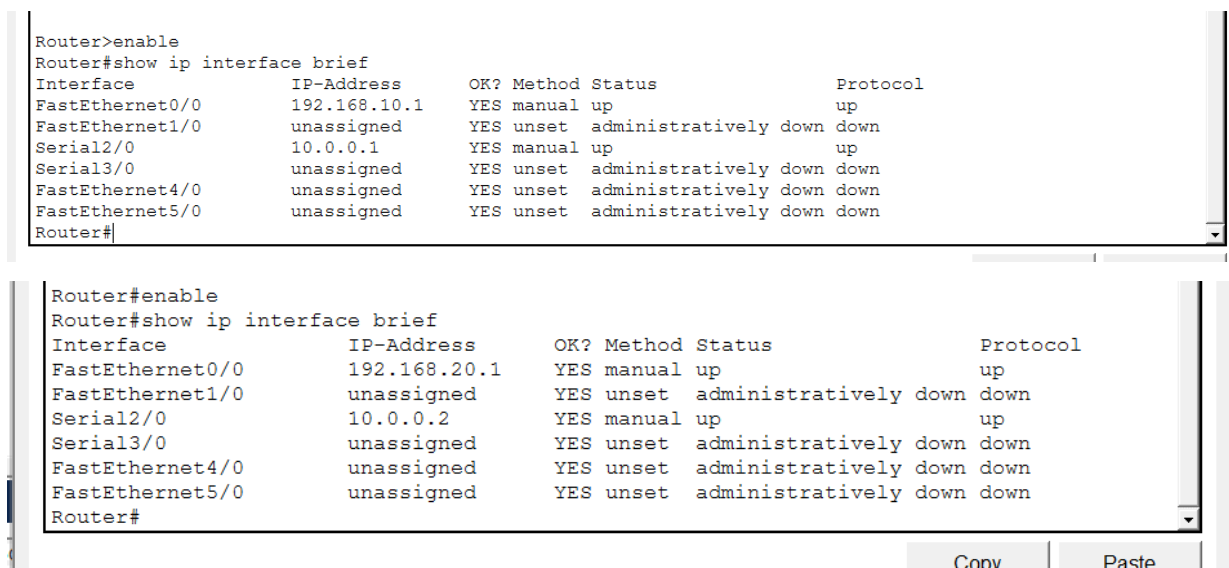


Figure 6: Verification of the R1–R2 serial link

5. OSPF Configuration

OSPF was configured to allow R1 and R2 to exchange routing information dynamically.

5.1 OSPF Configuration on R1

router ospf 1

network 192.168.10.0 0.0.0.255 area 0

network 10.0.0.0 0.0.0.3 area 0

```
router ospf 1
 log-adjacency-changes
 network 192.168.10.0 0.0.0.255 area 0
 network 10.0.0.0 0.0.0.3 area 0
```

Figure 7: OSPF configuration on R1

5.2 OSPF Configuration on R2

```
router ospf 1
```

```
network 192.168.20.0 0.0.0.255 area 0
```

```
network 10.0.0.0 0.0.0.3 area 0
```

```
router ospf 1
log-adjacency-changes
network 192.168.20.0 0.0.0.255 area 0
network 10.0.0.0 0.0.0.3 area 0
```

Figure 8: OSPF configuration on R2

6. OSPF Verification

The following command was used to check the OSPF neighbor relationship:

```
show ip ospf neighbor
```

The result showed:

Neighbor ID	State	Address
192.168.20.1	FULL	10.0.0.2

The **FULL** state confirms that R1 and R2 successfully established an OSPF neighbor relationship.

```
Router#show ip ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
192.168.20.1	0	FULL/ -	00:00:37	10.0.0.2	Serial2/0

```
Router#show ip ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
192.168.10.1	0	FULL/ -	00:00:39	10.0.0.1	Serial2/0

Figure 9: OSPF neighbor relationship between R1 and R2

7. Routing Table Verification

The routing table on R1 showed the remote network through OSPF:

O 192.168.20.0/24 via 10.0.0.2

```
Router#show ip route
```

Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route

Gateway of last resort is not set

```
10.0.0.0/30 is subnetted, 1 subnets
C    10.0.0.0 is directly connected, Serial2/0
C    192.168.10.0/24 is directly connected, FastEthernet0/0
O    192.168.20.0/24 [110/65] via 10.0.0.2, 00:37:04, Serial2/0
```

Figure 10: R1 routing table showing the OSPF learned route

The routing table on R2 showed:

O 192.168.10.0/24 via 10.0.0.1

```
Router#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

10.0.0.0/30 is subnetted, 1 subnets
C      10.0.0.0 is directly connected, Serial2/0
O      192.168.10.0/24 [110/65] via 10.0.0.1, 00:38:28, Serial2/0
C      192.168.20.0/24 is directly connected, FastEthernet0/0
```

Figure 11: R2 routing table showing the OSPF learned route

The letter **O** indicates that the route was learned through OSPF.

8. Connectivity Testing

Connectivity was tested between the PCs using the ping command.

For example, PC1 was used to ping PC3:

ping 192.168.20.10

The test successfully received replies from PC3.

The first test had one timeout, followed by successful replies. This can occur when the network is initially learning the required address information.

Further connectivity testing confirmed communication between the two LAN networks.

```
C:\>ping 192.168.20.10

Pinging 192.168.20.10 with 32 bytes of data:

Reply from 192.168.20.10: bytes=32 time=11ms TTL=126
Reply from 192.168.20.10: bytes=32 time=6ms TTL=126
Reply from 192.168.20.10: bytes=32 time=12ms TTL=126
Reply from 192.168.20.10: bytes=32 time=5ms TTL=126

Ping statistics for 192.168.20.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 5ms, Maximum = 12ms, Average = 8ms
```

Figure 12: Successful connectivity test from PC1 to PC3

```
C:\>ping 192.168.10.10

Pinging 192.168.10.10 with 32 bytes of data:

Reply from 192.168.10.10: bytes=32 time=10ms TTL=126
Reply from 192.168.10.10: bytes=32 time=11ms TTL=126
Reply from 192.168.10.10: bytes=32 time=2ms TTL=126
Reply from 192.168.10.10: bytes=32 time=1ms TTL=126

Ping statistics for 192.168.10.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 11ms, Average = 6ms
```

Figure 13: Successful connectivity test from PC3 to PC1

9. Saving the Configuration

After completing the configuration, the router configurations were saved using:

copy running-config startup-config

The configuration was saved successfully on both R1 and R2.

```
Router#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
```

Figure 14: Successful saving of router configuration

10. Conclusion

The network was successfully implemented in Cisco Packet Tracer. IP addresses were configured on the PCs and router interfaces, the serial connection between R1 and R2 was established, and OSPF was configured in area 0.

The OSPF neighbor relationship reached the **FULL** state, and both routers successfully learned the remote LAN through OSPF. Connectivity testing also demonstrated communication between devices on the two different LAN networks.

Therefore, the basic network implementation and dynamic routing configuration were successfully completed.

RUBRICS FOR CALCULATION

Criteria	Weightage	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
Problem Analysis	15%	Clearly identifies and deeply analyzes the industry problem with all factors	Good understanding with minor gaps	Basic understanding; some key aspects missing	Poor or incorrect understanding of the problem
Solution Design	20%	Innovative, practical, and industry-relevant solution	Effective solution with minor limitations	Partially suitable solution	Inappropriate or unclear solution
Technical Implementation	20%	Fully functional, accurate, and well-executed implementation	Mostly functional with minor errors	Partially implemented solution	Incomplete or non-functional implementation
Use of Tools & Technologies	10%	Appropriate and advanced use of relevant tools and technologies	Correct use of tools	Limited or basic use of tools	Incorrect or no use of tools
Problem-Solving Approach	10%	Logical, systematic, and well-structured approach	Mostly logical approach	Somewhat disorganized approach	No clear approach
Innovation & Creativity	10%	Highly creative and original solution	Some creativity	Limited originality	No creativity
Testing & Validation	5%	Thorough testing with real-world scenarios	Adequate testing	Minimal testing	No testing
Documentation & Presentation	5%	Clear, professional, and well-documented	Good documentation	Basic documentation	Poor or no documentation
Teamwork / Collaboration	5%	Excellent coordination and contribution (if team-based)	Good collaboration	Limited participation	No collaboration or contribution